# Shashank Subba

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## Education

**University of Georgia** **Expected May 2028**
*Bachelor of Science in Mechanical Engineering* | ***3.60 GPA*** *Athens, Georgia*
**Relevant Coursework:** Dynamics, Strength of Materials, Statics, Thermodynamics, Differential Equations, Linear Algebra

## Skills

**CAD:** SolidWorks, Creo Parametric, Fusion 360, Onshape, Inventor
**Mechanical & Fabrication:** Hardware Integration, Mechanical Assembly, GD&T, CNC Machining, FDM 3D Printing
**Analysis & Programming:** ANSYS Mechanical, MATLAB, Python, C++, Simulink

## Experience

**Quantum Space** May 2026 – August 2026
*Mechatronics Intern* *Rockville, MD*

- Led assembly, integration, and verification of 4 full-scale spacecraft prototype panels; coordinated installation of 10+ 3D-printed components with engineers and technicians, catching 6 fit issues before final assembly.
- Produced 8 fabrication-ready GD&T drawings for propulsion wire-clamp assemblies; modeled missing components and specified fabrication and installation tolerances across 6 harnesses.
- Fabricated a prototype optomechanical device; used Creo tolerance analysis to reduce misalignment to under 0.1 mm.
- Rebuilt a 100+ component Creo fastener and backshell library; reduced CAD insertion time by 40% companywide.
- Integrated AR-guided wire-harness routing using Magic Leap 2, Unity, and Vuforia, reducing assembly time by 13%.

**Smart Manufacturing and Robotics Lab** October 2026 – Present
*Robotics Integration Researcher* *Athens, GA*

- Developing a camera-controlled robotic arm and hand to replicate human movements, supporting the lab's VR-based manufacturing robotics research.
- Modeling printable finger mechanisms in SolidWorks and developing servo-driven cable actuation concepts for finger and wrist motion.

**Cooperative Automated Mobility Innovations Lab** December 2025 – May 2026
*Perception Systems Researcher* *Athens, GA*

- Validated 1,200+ autonomous-driving scenarios; identified safety-relevant vision-language model (VLM) errors in 40% of reviewed cases to support training-data corrections.
- Expanded Python/C++ diagnostic tools to check omitted object categories across 80+ autonomous-driving scenarios, extending validation coverage.

## Engineering Projects

**MicroMouse Competition Team | IEEE** August 2026 – Present
*Responsible Engineer, Mechanical Systems* *Athens, GA*

- Designing a 3.5:1 robot drivetrain in Fusion 360 within a 75 mm chassis; evaluating motor torque-speed behavior and 25 mm wheel clearance with 20T/70T, module 0.3 spur gears at 13.5 mm center distance.
- CNC-machined a maze test-section prototype from CAD; caught a 0.5 mm wall-slot tolerance error through dimensional inspection and fit checks before full-scale fabrication.

**NASA Human Exploration Rover Challenge | ASME** August 2025 – May 2026
*Drivetrain Engineer* *Athens, GA*

- Diagnosed shock loading behind chassis-support failure at approximately 40% of predicted torque capacity; analyzed redesigned supports in ANSYS Mechanical for a calculated safety factor of 2.3 under revised loads.
- Modeled a rover drivetrain assembly in Fusion 360 and produced GD&T drawings for 6 machined components, supporting fabrication, mechanical integration, and design review.

**CURO Assistive Navigation Glove** February 2026 – May 2026
*Mechanical Design Lead* *Athens, GA*

- Diagnosed motor-wire detachment in a wearable navigation glove; reinforced connections with heat-shrink and verified zero failures across 15+ wear/removal cycles with all 3 motors connected.
- Reduced cue misidentification by 75% through blindfolded testing; identified motor interference and increased spacing from 15 mm to 30 mm.